BROKERMASTER loader – deployment (RHEL 6.10)

Run as `root` unless a step says otherwise. Deploy to UAT (`NC100DITDEVAPPUAT`) first, then repeat for PROD.

0. Before you start

Item	Who	How
SQL login for the loader	DBA	`GRANT SELECT, INSERT, UPDATE ON dbo.BROKERMASTER TO [BROKER_LOADER];`
Backup of the table	DBA	`SELECT * INTO dbo.BROKERMASTER_BAK_20261014 FROM dbo.BROKERMASTER;`
New `DATEMODIFIED` column (after the backup)	DBA	`ALTER TABLE dbo.BROKERMASTER ADD DATEMODIFIED DATETIME NULL;` Existing rows stay NULL. First confirm no other code inserts into this table without a column list
Firewall: app server → SQL Server port (1433)	Network	Tested in step 6
Current delete/reload job identified	Load owner	It is turned off in step 10
JDBC driver downloaded on your workstation	You	`https://repo1.maven.org/maven2/com/microsoft/sqlserver/mssql-jdbc/12.8.1.jre8/mssql-jdbc-12.8.1.jre8.jar`

Check the driver on your workstation (PowerShell): `Get-FileHash .\mssql-jdbc-12.8.1.jre8.jar` must be `44F217B1D67D22442002EE2D1C4AB8E86358606A787B95EEB5301F235482A2AB`.

1. Create the service account

```
useradd -m -s /bin/bash mosetl
```

2. Create folders

```
mkdir -p /opt/mos/brokermaster/{classes,lib,archive} /var/log/mos
chown -R mosetl:mosetl /opt/mos/brokermaster /var/log/mos
chmod 750 /opt/mos/brokermaster
```

3. Copy the files to the server

With WinSCP (or `scp`), copy into `/opt/mos/brokermaster/`:

File	Destination
`run_brokermaster.sh`	`/opt/mos/brokermaster/`
`.env.example`	`/opt/mos/brokermaster/` (rename if the download dropped the leading dot)
`brokermaster-loader.jar` (pre-compiled for Java 8)	`/opt/mos/brokermaster/lib/`
`mssql-jdbc-12.8.1.jre8.jar`	`/opt/mos/brokermaster/lib/`

Then fix Windows line endings and ownership, and check both jars:

```
cd /opt/mos/brokermaster
sed -i 's/\r$//' run_brokermaster.sh .env.example
chown -R mosetl:mosetl /opt/mos/brokermaster
chmod 750 run_brokermaster.sh
sha256sum lib/*.jar
```

Jar	SHA-256
`brokermaster-loader.jar`	`80ba1fb2d4474ab889167326561abcfeac1b87366897807f0a5c871eb332b35a`
`mssql-jdbc-12.8.1.jre8.jar`	`44f217b1d67d22442002ee2d1c4ab8e86358606a787b95eeb5301f235482a2ab`

4. Check Java 8

Nothing to compile on the server: the loader jar is already built for Java 8 (`javac` is not executable on this server).

On this server `/usr/java/latest` and `/usr/java/default` point to **Java 6** and the default `java` is Java 7. Neither can run the loader. The Java 8 runtimes are:

```
ls -l /usr/java/jdk1.8.0_291-amd64/jre/bin/java /usr/java/jdk1.8.0_291-amd64/bin/java
/usr/java/jre1.8.0_281/bin/java
su - mosetl -c '/usr/java/jdk1.8.0_291-amd64/jre/bin/java -version'   # expect java version "1.8.0_291"
su - mosetl -c '/usr/java/jre1.8.0_281/bin/java -version'             # expect java version "1.8.0_281"
```

At least one must print a `1.8.0_…` version. `run_brokermaster.sh` checks them in that order, uses the first that runs and reports 1.8, and never falls back to Java 6/7. If you want to pin one, set it as `JAVA_BIN` in `.env` (step 5). If all give `Permission denied`, ask the Linux team for execute access to a Java 8 runtime.

5. Create `.env`

```
cd /opt/mos/brokermaster
cp .env.example .env
chown mosetl:mosetl .env && chmod 600 .env
vi .env
```

Set (keep the single quotes):

Setting	Value
`DB_URL`	`'jdbc:sqlserver://<server>:1433;databaseName=<database>;encrypt=true;trustServerCertificate=true'`
`DB_USER` / `DB_PASSWORD`	The SQL login from step 0
`BROKER_TABLE`	`'dbo.BROKERMASTER'`
`BROKER_ID_MODE`	`'identity'` (confirmed: `ID` is IDENTITY)
`BROKER_MODIFIED_COLUMN`	`'DATEMODIFIED'`, only once the column exists (step 0). Leave empty to turn the feature off
`BROKER_ARCHIVE_DIR` / `BROKER_STATE_FILE`	Leave as in the example (`/opt/mos/brokermaster/...`)

6. Check access

```
# Source file readable by the service account
su - mosetl -c 'test -r "/usr/local/mos/brokerinformation/Master Broker Channel List.XLSX" && echo FILE OK'

# SQL Server port reachable (replace host)
timeout 5 bash -c '</dev/tcp/<server>/1433' && echo PORT OK
```

If the file check fails: `ls -l /usr/local/mos/brokerinformation/` and give `mosetl` read access, e.g. add it to the folder's group (`usermod -a -G <group> mosetl`) and make sure the file is group-readable.

7. Dry run (changes nothing)

```
su - mosetl -c '/opt/mos/brokermaster/run_brokermaster.sh --dry-run --force; echo "exit=$?"'
tail -20 /var/log/mos/brokermaster_load.log
ls -t /var/log/mos/brokermaster_exceptions_*.csv | head -1
```

Check	Expect
Exit code	`0`, or `2` if rows went to the exceptions CSV
`Read ...: N unique CPSCODE rows`	About 1,550
`section heading 'REMOVED FROM PMPP' skipped`	Present
`DRY RUN (rolled back) ... inserted=, updated=, unchanged=`	`inserted` small; `unchanged + updated` ≈ rows in table
`in_db_not_in_file`	Close to 0
Exceptions CSV	Review the duplicates and rejected rows

Stop here if the numbers look wrong; nothing has been changed.

8. First real run

```
su - mosetl -c '/opt/mos/brokermaster/run_brokermaster.sh --force; echo "exit=$?"'
tail -5 /var/log/mos/brokermaster_load.log
```

Verify in SQL:

```
SELECT COUNT(*) AS total_rows, COUNT(DISTINCT CPSCODE) AS unique_codes FROM dbo.BROKERMASTER;  -- must be equal
SELECT TOP 20 * FROM dbo.BROKERMASTER ORDER BY ID DESC;                                        -- newly inserted rows
SELECT DATEMODIFIED, COUNT(*) AS row_count FROM dbo.BROKERMASTER GROUP BY DATEMODIFIED ORDER BY DATEMODIFIED DESC;
-- one timestamp per run that changed rows; NULL = not changed since tracking started
```

Run it again without `--force`: the log should say `unchanged since last load; nothing to do`.

9. Schedule

```
crontab -u mosetl -e
```

Add (weekdays at 6:15 AM; change the time to suit):

```
MAILTO=your.email@bank.com
15 6 * * 1-5  /opt/mos/brokermaster/run_brokermaster.sh
```

Cron only emails when there are warnings or errors. Confirm the server can send external mail (`echo test | mail -s test your.email@bank.com`); if not, check the log daily or ask the Linux team to set up a mail relay.

10. Turn off the old process

Disable the existing delete/reload job so the two loads don't overwrite each other.

11. Log rotation

```
cat > /etc/logrotate.d/brokermaster <<'EOF'
/var/log/mos/brokermaster_load.log {
    monthly
    rotate 12
    compress
    missingok
    notifempty
    create 0640 mosetl mosetl
}
EOF
```

Optional: clean up old exceptions CSVs and archive copies by adding this line to the same crontab:

```
30 6 * * 0  find /var/log/mos -name 'brokermaster_exceptions_*.csv' -mtime +90 -delete; find
/opt/mos/brokermaster/archive -mtime +365 -delete
```

Rollback

Step	Command
Stop the job	`crontab -u mosetl -e` and comment out the line
Restore data (DBA)	Restore `dbo.BROKERMASTER` from `dbo.BROKERMASTER_BAK_20261014`
Re-enable the old process	As before

Troubleshooting

Error in the log	Fix
`UnsupportedClassVersionError`	Ran with Java 7. Set `JAVA_BIN` in `.env` to the Java 8 path
`Permission denied` running `java`	See step 4
`Could not find or load main class BrokerMasterLoader`	`brokermaster-loader.jar` is missing from `lib/`
`/bin/bash^M: bad interpreter`	Windows line endings: rerun the `sed` command in step 3
`DB_URL is not set`	`.env` missing or not readable by `mosetl`
`The TCP/IP connection to the host ... has failed`	Wrong server/port, or firewall. Recheck step 6
`Login failed for user`	Wrong `DB_USER` / `DB_PASSWORD`, or SQL authentication disabled on the server
`SSL` / `TLS` / `PKIX` errors	Keep `trustServerCertificate=true`, or ask the DBA whether the server supports TLS 1.2
`Missing column(s)`	A heading in the Excel file changed. The error lists the headings found
`appears more than once (check hidden columns)`	Duplicate heading, often in hidden columns J–P
`already has duplicate CPSCODE`	Clean up duplicates in `BROKERMASTER` first
`already has duplicate CPSCODE` but the table looks empty in DBeaver	The loader is connected elsewhere. Compare the log's `Connected to server=… database=…` with `SELECT @@SERVERNAME, DB_NAME()` in DBeaver. For a named instance (`HOST\INSTANCE`), put that instance's port in `DB_URL`, or use `;instanceName=…`
Job never runs	`grep CROND /var/log/cron`; check `crontab -u mosetl -l`